The Compiled Core

From Generative AI to Deterministic Infrastructure for Regulatory Reasoning in Organ Procurement

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Autonomous OPO Reasoning and Triage Architecture

> *The purpose of abstraction is not to be vague, but to create a new semantic level in which one can be absolutely precise.*

— Edsger W. Dijkstra

> *He didn't compute the inverse square root faster. He discovered that the answer was already latent in the representation.*

— on John Carmack's fast inverse square root, Quake III Arena (1999)

Abstract

Organ procurement organizations face mounting pressure to adopt artificial intelligence under CMS final rule CMS-3409-P. This paper synthesizes the complete AORTA research program into a single architecture and identifies its deepest structural insight: **the reasoning methodology this program has developed is not input to a language model—it is source code for a deterministic system that can be compiled and executed without a language model for the vast majority of operational queries.**

The architecture presented here treats the large language model as a *compiler*—an expensive, offline tool that manufactures reasoning artifacts—rather than as a *runtime* that must be invoked on every query. The manufactured artifacts (a compositional grammar of typed reasoning primitives and a validated dataset of reasoning traces) are the compiled output. A three-tier runtime dispatches queries to the appropriate execution mode: fully compiled deterministic lookup for routine queries (70–80% of volume), grammar-guided constrained generation for two-authority intersections (15–25%), and full compositional reasoning for genuinely novel scenarios (5–10%).

The compiled tier requires no language model, no API, no GPU, and no internet connection. It runs on commodity hardware in microseconds. It produces zero hallucinations because it performs no generation—only retrieval and assembly of pre-validated components. It is fully auditable by non-technical stakeholders, including CMS surveyors, because every component is human-readable and deterministic. It sidesteps the Software as Medical Device regulatory classification because it is not a predictive algorithm—it is a lookup table assembled by human-validated templates.

The framework is portable across models, organizations, and regulatory updates. It is released under open license. This paper specifies the complete architecture, maps every prior publication in the research program to its role, provides the implementation blueprint, and establishes the empirical validation agenda required to convert design into evidence.

Part I: The Problem

1. The Coordinator at 3 AM

At 3:14 AM, a transplant coordinator receives notification of a potential organ donor. Within the next several hours, she must navigate a regulatory environment spanning federal statute (the National Organ Transplant Act), CMS conditions of participation (42 CFR 486 Subpart G), Organ Procurement and Transplantation Network policies, her state's Uniform Anatomical Gift Act, and her organization's internal standard operating procedures. Many of these authorities address overlapping domains. Some conflict. The consequences of error range from regulatory citation to organ loss to patient death.

She does not need an AI system that retrieves the right policy text. She needs a system that reasons correctly about how multiple regulatory authorities intersect in the specific operational scenario she faces—and that knows what it must not attempt to decide.

CMS final rule CMS-3409-P, published January 30, 2026, makes this need urgent. The rule establishes a three-tier performance evaluation system with decertification consequences for the 56 OPOs operating in the United States. Regulatory compliance and regulatory optimization are no longer synonymous: a coordinator can be fully compliant with OPTN allocation policy while making decisions that damage her OPO's CMS performance tier. The rule has a phased implementation timeline that creates a complex temporal space requiring careful phase-aware reasoning.

2. The Reasoning Pattern Gap

Retrieval-augmented generation addresses the knowledge retrieval problem: given a question about OPTN Policy 2.6, the system retrieves the relevant policy sections and provides an accurate response. RAG does not address the **reasoning pattern problem**. When a query requires synthesizing information across multiple regulatory authorities—determining jurisdictional primacy, resolving apparent conflicts, accounting for temporal status of proposed rules—the model must perform multi-hop reasoning that retrieved text chunks alone do not scaffold.

This gap manifests in six characteristic failure modes:

Failure Mode	Description	Frequency
Jurisdictional misattribution	Applies the wrong authority to a given scenario	Most common, most dangerous

Temporal conflation	Treats proposed rules as binding or fails to distinguish regulatory phases	Acute under CMS-3409-P
Hierarchy collapse	Confuses specificity with superiority across authority tiers	Common at intersections
Gap blindness	Generates confident answers where no authority provides clear guidance	High confidence, high risk
Boundary violation	Provides clinical determinations exceeding AI scope	Catastrophic when occurs
Metric blindness	Fails to account for CMS-3409-P performance tier impacts	New under final rule

These failures cluster at regulatory intersections—the seams where two or more authorities overlap. An inverse relationship obtains between intersection frequency and failure consequence: multi-authority intersections are rare (5–10% of coordinator queries) but carry disproportionate risk because they represent exactly the scenarios where unscaffolded models fail most dangerously, often with high confidence.

3. The Minecraft CPU Problem

The standard approach to applying AI in organ procurement follows a pattern common across industry:

Component	What It Does
LLM as reasoning engine	Processes natural language queries about the domain
RAG retrieval	Fetches relevant documents to inform responses
Agent frameworks	Orchestrate multiple LLM calls with tool use
Fine-tuning	Adapts the model to domain-specific language
Wrapper applications	UI and integration connecting LLM to existing systems

This architecture works. Value is created. But notice what is happening: the AI system is being used to *simulate* reasoning about the problem at inference time. The actual decision logic—authority identification, temporal resolution, hierarchy application, conflict adjudication—is being reconstructed through language processing on every query.

In the game Minecraft, players have built functional computers using in-game redstone components. These constructions work. They are also absurd: a Minecraft CPU executes one instruction per minute while the physical computer running Minecraft executes billions per second. The player is simulating computation on top of a substrate that already computes, throwing away multiple orders of magnitude of capability.

Using a language model to simulate regulatory reasoning at inference time, when that reasoning is fully specifiable as a deterministic procedure, is building a CPU inside Minecraft. The language model is the wrong substrate for the majority of this problem. The question is what the right substrate is.

Part II: The Architecture

4. The Core Insight: Reasoning as Compilation

In 1999, John Carmack's fast inverse square root function achieved a categorical speedup not by computing the answer faster, but by discovering that the answer was already latent in the representation. The IEEE 754 floating-point format encodes numbers such that their bit pattern, reinterpreted as an integer, approximates the logarithm. The magic constant 0x5f3759df aligns the bits; a single Newton-Raphson iteration refines the approximation. What had required expensive floating-point operations became a cast, a subtraction, a bit shift, and one multiply-add.

The central insight of this paper is that the same structural opportunity exists in regulatory AI alignment. The decision logic of organ procurement regulatory reasoning—which authority governs, how authorities relate hierarchically, how temporal phases interact, where genuine gaps exist—is not a reasoning problem for the majority of operational queries. **It is a classification and lookup problem.** The reasoning has already been done. The answers are already latent in the regulatory structure itself. The task is not to compute them at inference time but to compile them ahead of time and look them up at runtime.

The language model's role is not to perform this reasoning in production. Its role is to *manufacture* the compiled artifacts during an offline generation pipeline—to act as a compiler that translates regulatory logic into deterministic executable components. Once the compilation is complete, the LLM becomes unnecessary for the compiled tier, just as ENIAC became unnecessary once the artillery tables were printed.

4.1 The Compiler Metaphor Made Precise

Compiler Concept	AORTA Equivalent
Source language	Regulatory logic (CMS-3409-P, OPTN policies, UAGA, NOTA)
Compiler	The multi-agent generation pipeline (offline, expensive, run once)
Intermediate representation	The bimodal core (compositional grammar + reasoning trace dataset)
Target architectures	Tier 1 (deterministic FSM), Tier 2 (constrained LLM), Tier 3 (full LLM)
Lexical analysis	Query canonicalization (raw coordinator input → structured form)
Semantic analysis	Authority router (canonical form → intersection signature)

Code generation	Grammar composition (intersection signature → primitive chain)
Executable binary	Pre-assembled response templates in the trace library
Runtime	Three-tier dispatcher

4.2 The Magic Constant

Carmack's magic constant worked because IEEE 754 floats are already logarithms in disguise. The bits encode the structure of the problem; the constant just aligns them correctly.

The authority router plays an analogous role. It recognizes that regulatory authority forms a precedence-ordered set, that jurisdictional conflicts form a computable lattice, and that coordinator queries map to a finite set of intersection signatures. The router is a hash function that collapses the infinite-dimensional space of human language into discrete coordinates in the combinatorial trace matrix:

```
itinerary = authority_router.route(query, case_context)
# Output: { signature: 'UAGA+OPTN+SOP', complexity: 'edge', jurisdiction: 'TX'
}
# This signature is a memory address in the compiled binary.
# The reasoning at that address is already done.
```

Once hashed, lookup is O(1). For single-authority queries—which constitute 70–80% of coordinator interactions—the router already knows exactly which authority governs. The answer is retrieval and formatting, not generation. No language model required.

5. The Bimodal Core

The compiled output of the generation pipeline consists of two inseparable artifacts that together form the bimodal core—the central deliverable of the AORTA research program.

5.1 Artifact 1: The Compositional Reasoning Grammar

A formal system comprising eight typed reasoning primitives, each addressing a specific failure mode from the reasoning pattern gap. The primitives compose according to explicit rules determined by the intersection signature. The grammar is organized within a seven-layer architecture: a deterministic pre-processing substrate, the compositional grammar itself, a safety automaton, an inference engine, an audit-grade output contract, a continuous evolution loop, and a proactive navigation layer.

Three properties distinguish this grammar from standard approaches. First, authority identification is resolved deterministically before the language model engages, eliminating the most common failure class without spending a single language model token. Second, safety boundaries are enforced by automaton topology—there is no path from any state to output that bypasses the boundary enforcement check. The Human Line is enforced by the absence of a graph edge, not by a prompt instruction. Third, the grammar generates correct reasoning chains for regulatory intersections the system has never encountered, because composition from typed primitives is generative.

Primitive Types

Primitive	Addresses	Tokens	Position
HIERARCHY_RESOLVE	Hierarchy collapse	150–300	After TEMPORAL
TEMPORAL_CLASSIFY	Temporal conflation	100–200	Always first
CONFLICT_ADJUDICATE	Jurisdictional misattribution	200–400	After TEMPORAL
GAP_DETECT	Gap blindness	150–250	After HIERARCHY
BOUNDARY_ENFORCE	Boundary violation	100–200	Before OUTPUT
CASCADE_MAP	Metric blindness	200–400	After SYNTHESIZE
SYNTHESIZE	Final assembly	200–400	After resolution

CALIBRATE	Confidence assignment	100–150	Final step

Composition Rules

Single-authority: T → S → M → B → K. Two-authority: $T(a_1) \rightarrow T(a_2) \rightarrow H(a_1,a_2)$ → S → M → B → K. Two-authority with conflict: $T(a_1) \rightarrow T(a_2) \rightarrow C(a_1,a_2)$ → H → S → M → B → K. Three or more authorities: [$T(a_n)$ for all] → [$C(a_i,a_j)$ for conflicting pairs] → [H for each] → G → S → M → B → K. A compliance officer can read this grammar and verify it in an afternoon.

Counter-Traces

For each high-risk primitive, the library includes a paired counter-trace: a compressed demonstration of the wrong reasoning pattern paired with forensic diagnosis. Total cost: approximately 500 tokens. This is the highest-ROI token expenditure in the system. Each counter-trace provides targeted immunization against the specific failure mode most likely to occur at its corresponding intersection.

5.2 Artifact 2: The Reasoning Trace Dataset

A structured library of complete regulatory reasoning chains produced by executing the grammar against archetypal scenarios, validated through a multi-stage pipeline, and organized in four layers:

Layer 0 — Behavioral Core. Confidence calibration, anti-sycophancy, Human Line enforcement, appropriate refusal. A few hundred examples defining how the model behaves. Universal across all OPOs.

Layer 1 — Regulatory Foundation. Multi-step reasoning traces over the shared policy landscape: OPTN allocation, CMS conditions, federal regulations. Large, universally applicable, directly validatable against public documents.

Layer 2 — Clinical Reasoning. Decision traces at graduated complexity. Tier A: standard single-organ referrals. Tier B: multi-organ standard donors. Tier C: complex scenarios (DCD, infectious disease, pediatric). Tier D: edge cases (policy contradictions, novel presentations, genuine gaps). Enables progressive adoption.

Layer 3 — Regional Adaptation Templates. How state-level UAGA provisions integrate into reasoning chains. Intentionally incomplete—each OPO customizes for their jurisdiction.

Triple-Use Format

Each trace is stored as a structured object serving three deployment modes from a single artifact: **context injection** (frontier API models—scenario and reasoning chain injected into system prompt or retrieved dynamically), **fine-tuning data** (open-weight local models—scenario becomes user turn, reasoning chain becomes assistant turn), and **evaluation benchmarking** (scenario as test item, trace as gold standard for AORTA-Bench). A single generation effort produces three deployment-ready artifacts.

5.3 The Generative Relationship

The grammar and the dataset are not alternative approaches. They are related by a generative function: the grammar defines the methodology; when executed against scenarios, it produces traces. The grammar generates the dataset. The dataset validates the grammar. If traces consistently pass safety gates and domain experts confirm correctness, the grammar is validated. If traces fail, failures point to specific grammar components requiring revision.

Together, the grammar handles novel scenarios through composition; the dataset handles familiar scenarios through retrieval. The grammar provides generative methodology; the dataset provides empirical demonstrations. Neither is complete without the other.

6. The Three-Tier Runtime

The three-tier runtime is the execution architecture for the compiled bimodal core. It dispatches each query to the minimum computational tier required, achieving near-zero latency for the majority of operations while preserving full reasoning capability for the hard tail.

6.1 The Memory Hierarchy Analogy

The three tiers map precisely to a computer's memory hierarchy, with a critical safety implication: the unpredictable substrate (the language model) is isolated to the slowest, most expensive, least-used tier.

Tier	Analog	Latency	Safety	Volume	Hallucination
Tier 1	L1 Cache	Microseconds	Deterministic	70–80%	Zero
Tier 2	L2 Cache	Seconds	Constrained	15–25%	Bounded
Tier 3	Main Memory	Seconds–minutes	Generative	5–10%	Possible

6.2 Tier 1: The Compiled Tier

For single-authority queries, the authority router identifies the governing authority with high confidence. The trace library contains a pre-validated response template for this authority-intent combination. The output contract assembler populates the nine-field schema from the template and retrieved policy chunks.

```
def respond_tier1(query, case_context):
    itinerary = authority_router.route(query, case_context)
    trace = trace_library.get(itinerary.signature, query.intent)
    chunks = policy_index.retrieve(itinerary.primary_authority, query)
    return output_contract.render(trace, chunks, itinerary)
```

This function contains no language model call, no stochastic generation, no prompt engineering. It is pure deterministic logic. It runs in microseconds. It is fully auditable. It can be embedded directly into an OPO's case management system as a local library requiring no internet connection, no API keys, and no recurring costs. Hardware requirement: any computer manufactured in the last decade.

6.3 Tier 2: Grammar-Guided Generation

For two-authority intersections where the authority pair is known but the specific factual configuration may be novel, the grammar selects the primitive chain and the model instantiates it against the specific facts. The language model is not reasoning from first principles—it is executing a pre-specified methodology against a novel combination of variables. The safety automaton ensures the output respects the Human Line and follows the required state traversal.

Tier 2 queries are candidates for promotion to Tier 1. Each time a Tier 2 response is accepted by a coordinator, it becomes a candidate for a new trace. Over time, as the trace library grows, the compiled proportion increases. The system becomes more deterministic—and therefore safer—through use.

6.4 Tier 3: Full Compositional Reasoning

For genuinely novel intersections—three or more authorities, unprecedented regulatory developments, scenarios where the grammar must compose primitives for combinations never encountered—the frontier model receives the full grammar in context and performs compositional reasoning. Human escalation is already mandated for these cases by operational protocol. The AI provides structured analysis; the coordinator and supervisor make the decision.

Tier 3 queries are the raw material for the continuous evolution pipeline. Every resolved Tier 3 query, once validated, generates a new trace that may eventually promote to Tier 2 or Tier 1. The system consumes AI capability to produce deterministic safety.

6.5 The Self-Obsoleting Property

The architecture has a thermodynamic property: it cools over time. When a new regulatory development occurs (CMS-3409-P, a novel OPTN policy), the domain is "hot"—high uncertainty, heavy reliance on Tier 3. As traces are generated, validated, and compiled into the library, the system cools. Uncertainty is replaced by compiled structure. The Tier 3 proportion decreases. The Tier 1 proportion increases.

The asymptotic end-state is a system where the entire combinatorial space of regulatory intersections has been compiled into deterministic components. The language model becomes vestigial—a bootstrap compiler that has completed its job.

This is the opposite of every SaaS AI business model, which seeks to increase model dependence over time. This architecture is explicitly designed to eliminate its own need to use AI. And that elimination is precisely what makes it trustworthy in healthcare: the system gets *safer* over time because it gets more deterministic.

7. The Safety Architecture

7.1 The Safety Automaton

Every query traverses a state machine: INTAKE → AUTHORITY_ROUTE → TEMPORAL_CLASSIFY → [CONFLICT_TEST → HIERARCHY_RESOLVE if multi-authority; GAP_DETECT if irreconcilable] → SYNTHESIZE + CALIBRATE → CASCADE_MAP → BOUNDARY_ENFORCE → OUTPUT.

There is no path from any state to OUTPUT that bypasses BOUNDARY_ENFORCE. This is not a policy instruction. It is a structural property of the automaton's topology. The Human Line—the set of decisions that must remain with human practitioners regardless of AI capability—is enforced by the absence of a graph edge, not by a textual admonition. No amount of prompt injection, adversarial framing, or escalating scope creep can cause the system to produce a clinical determination, an eligibility assessment, or a family communication recommendation. The path through which such output would be generated does not exist in the automaton.

7.2 The SaMD Bypass

This is the single most politically consequential property of the architecture.

The FDA and CMS regulatory frameworks for Software as a Medical Device (SaMD) are designed to evaluate predictive algorithms—systems that take clinical inputs and produce clinical outputs through computational processes that cannot be inspected at the instance level. The Tier 1 compiled appliance does not meet this definition. It is not a predictive algorithm. It is a **deterministic lookup and assembly system** that retrieves pre-validated text written by human domain experts and formats it according to a fixed template.

When a CMS surveyor audits the system, they do not see model weights. They see the authority router source code (a deterministic decision tree), the trace library (human-validated text documents), and the assembly logic (template concatenation). This is *configuration*, not computation. It is medically safe by construction.

The AI component—the language model—exists only in the offline generation pipeline (where its outputs are validated by human experts before entering the library) and in the Tier 2/3 runtime (where human escalation is already mandated by operational protocol). The operational core of the system is firewalled from generative AI risk.

7.3 The Nine-Field Output Contract

Every response follows a fixed structure that makes the system audit-grade: (1) Regulatory state anchor with case date and applicable phase. (2) Authority itinerary with ordered authorities, status, and scope. (3) Determination—what the system can safely state, or DEFERRED. (4)

Regulatory cascade—downstream obligations triggered, with deadlines. (5) Performance metric impact—CMS-3409-P metrics affected, with explicit caveat. (6) Confidence level with justification. (7) Boundaries respected—explicit enumeration of what the response does NOT determine. (8) Escalation path. (9) Evidence pack—citations, primitive IDs, automaton path, itinerary hash.

7.4 The Human Line

The system never provides clinical viability determinations, eligibility assessments, transplant center clinical decision-making evaluations, or family communication recommendations. These boundaries are not textual instructions that the model must choose to follow. In Tier 1, they are absent from the templates by construction. In Tier 2 and 3, they are enforced by the automaton's topology. The coordinator's clinical judgment, empathy, and presence with the family are the core wire of organ procurement. The compiled appliance is the scaffolding that frees the coordinator to focus entirely on that wire.

8. Unified Deployment Model

8.1 The Access Tier Spectrum

Access Tier	Model Type	Primary Deliverable	Mechanism
Frontier API	Claude, GPT, Gemini	Grammar + traces in context window	No weight modification
Open-Weight Local	Qwen, LLaMA, Mistral	Trace dataset as fine-tuning data	QLoRA or full fine-tune
Hybrid	Frontier + local	Compiled tier for routine; API for edge cases	Complexity-based routing
Compiled Only	No LLM required	Tier 1 deterministic appliance	Lookup and assembly
Minimal	Any chat interface	Traces as few-shot examples	Manual context injection

The bimodal core covers every tier from a single pair of artifacts. No alternative architecture—a model-specific fine-tune, a proprietary product, or a context-only approach—provides this coverage breadth. The “Compiled Only” tier is new: an OPO with no AI infrastructure whatsoever can deploy the Tier 1 appliance as a standalone Python application.

8.2 Community Architecture

Layer A (Archetypal, portable, public) contains the universal grammar, trace library, counter-traces, automaton specification, and generation pipeline specification. Built against the shared regulatory core common to all 56 OPOs. Released under MIT license.

Layer B (Organization-specific, private) contains traces generated against a particular OPO’s internal SOPs, state UAGA, territorial boundaries, and CMS-3409-P tier position. Each OPO generates their own Layer B.

When one OPO discovers a failure mode and generates a counter-trace, it enters the shared Layer A library. When another encounters a novel intersection and generates new primitives, those primitives benefit all 56 organizations. Operational experience at any OPO strengthens the reasoning infrastructure for every OPO. This is herd immunity for regulatory reasoning.

9. The Generation Pipeline

The offline pipeline manufactures both artifacts of the bimodal core. The grammar defines what the pipeline builds; the traces are what it produces.

9.1 The Coordinator Ontology

The pipeline begins with a structured understanding of how coordinators actually reason, derived from practitioner interviews. The ontology captures decision skeletons (the order experienced coordinators process information), cognitive load maps (what is held in working memory simultaneously), confidence thresholds (when to seek human backup), and the Human Line as practitioners define it. The ontology shapes scenario generation: scenarios reflect actual decision architecture, not brute-force question-answer pairs from policy documents.

9.2 Five-Stage Pipeline

Stage 1: Scenario Generation. A frontier model generates scenarios targeting each cell in the combinatorial trace matrix, shaped by the coordinator ontology.

Stage 2: Grammar-Guided Reasoning. Each scenario is processed through the compositional grammar. For complex scenarios, multi-agent debate (Compliance Advocate, Utilization Advocate, Boundary Guardian) pressure-tests the reasoning from multiple angles.

Stage 3: Cross-Verification. Adversarial review by a separate model instance identifies errors, missed authorities, overconfident claims, or boundary violations. Counter-traces are generated at this stage.

Stage 4: Human Curation. Domain experts review cross-verified traces, correct remaining errors, validate composition, and select gold traces.

Stage 5: Deployment Formatting. Curated traces are formatted simultaneously for context injection, fine-tuning, and evaluation benchmarking.

9.3 Continuous Evolution

The pipeline operates continuously. Coordinator feedback feeds back into scenario generation. Expected Harm Scoring (frequency × consequence severity × observed uncertainty) prioritizes generation budget. Regulatory diff propagation triages affected traces when policy changes occur: approximately 80% temporal-only patches (automated, seconds), 15% content changes (light review, minutes), 5% pattern-affecting changes (full regeneration, hours per primitive). This represents approximately 95% reduction in staleness resolution cost versus regenerating complete demonstrations.

10. Relationship to Prior Publications

This paper synthesizes and extends four prior publications in the AORTA research program. Each addressed a specific facet of the problem; this paper identifies their structural unity.

Publication	Contribution	Role in Compiled Core
Reasoning Trace Injection (RTI)	Method for transferring reasoning via context window; trace format specification; combinatorial matrix	Defines Artifact 2 (the dataset) and its deployment for closed models
Reasoning as Infrastructure	Compositional grammar; typed primitives; safety automaton; seven-layer architecture; proactive navigation	Defines Artifact 1 (the grammar) and the safety architecture
The Bimodal Core	Unified grammar + dataset as central deliverable; hourglass architecture; deployment coverage analysis	Identifies the generative relationship between Artifacts 1 and 2
Native Intelligence	Substrate-native computation thesis; scaffolding ratio; Carmack/Chrono-Sieve precedents; Sinkhorn attention for allocation	Provides the theoretical basis for the compilation insight (Sections 3–4)

The structural insight that unifies these publications is that Native Intelligence asked the right question ("what if the problem IS the native operation?") but initially applied it to the wrong substrate (GPU operations for allocation). The bimodal core papers answered the question correctly: the native substrate for regulatory reasoning is not a neural network but a deterministic state machine, and the compositional grammar is its instruction set architecture.

11. Empirical Validation Agenda

Everything in this paper is architectural design. The critical next step is empirical evidence. The following validation agenda is designed to convert design into demonstrated capability.

11.1 The Decomposition Experiment

AORTA-Bench provides the evaluation instrument. The experiment measures the performance delta across five conditions:

#	Condition	What It Measures
1	Base model + RAG only	Baseline: how well does retrieval alone address the reasoning gap?
2	Base model + RAG + grammar	Grammar delta: how much does the compositional methodology add?
3	Base model + RAG + trace retrieval	Dataset delta: how much do reasoning demonstrations add?
4	Fine-tuned model on trace dataset	Weight embedding: does fine-tuning internalize the patterns?
5	Fine-tuned model + grammar overlay	Combined: does grammar add value even over fine-tuning?

Safety gates must be passed before capability is scored. A model that produces clinically unsafe output scores zero regardless of factual accuracy. The results will determine not just whether the framework works, but where its value concentrates—informing resource allocation for development.

11.2 The Compiled Tier Validation

Separately, the Tier 1 deterministic appliance must be validated against the same benchmark. For the single-authority queries it is designed to handle, does template assembly match or exceed the quality of LLM-generated responses? If yes, the compiled tier is validated. If no, the trace library needs refinement.

11.3 Coordinator Validation

Ten traces, shown to coordinators who work the core wire at 3 AM, with one question: “Does this reflect how you actually think?” If yes, the ontology is grounded. If no, the ontology’s errors are identified from the source that matters.

12. Future Research Directions

The compiled core architecture opens several research directions beyond the scope of this paper. These are noted briefly for completeness; detailed investigation is deferred to future work.

12.1 Near-Term Extensions

Shadow validation. Running Tier 2/3 in parallel with Tier 1 on a sampling basis, logging divergence deltas to detect concept drift, trace library errors, or regulatory changes not yet flagged. Uses the speed differential between tiers as a surveillance budget rather than a latency optimization.

Tier promotion automation. Systematic pipeline for promoting successfully resolved Tier 2 queries to Tier 1 traces, accelerating the thermodynamic cooling of the system through operational use.

Cross-domain portability. Clinical trial compliance (FDA/IRB intersections), healthcare billing (CMS/payer intersections), and environmental permitting (EPA/state agency intersections) share the same structural properties: bounded authority sets, enumerable intersections, reasoning failures at seams. The architecture is domain-agnostic; only the primitives are domain-specific.

12.2 Longer-Term Research

Formal verification. The grammar is small and explicit enough for encoding in TLA+ or Coq. Proving that no valid composition of primitives produces a clinical determination (the Human Line theorem) would convert the safety guarantee from structural to mathematical—a legal instrument, not just an architectural property.

Privacy-preserving audit. Because the authority router is deterministic and the intersection signature reveals nothing about the patient, cryptographic proof of correct routing is theoretically possible. A regulator could verify "this OPO followed the approved methodology" without seeing patient data. This resolves the tension between auditability and HIPAA compliance by design.

Distributed state management. Organ procurement is physically distributed across hospitals, highways, and homes. The obligation tracker (Layer 7) could be implemented as a conflict-free replicated data type (CRDT), allowing seamless case handoffs without central database dependency.

Substrate-native allocation. The Sinkhorn attention mechanism described in *Native Intelligence* remains a viable research direction for the distinct problem of donor-recipient matching (optimal transport), as opposed to the regulatory compliance problem addressed here. The two systems would operate at different layers: the compiled appliance handles regulatory reasoning; a substrate-native allocation engine would handle the matching computation itself.

13. Limitations

The framework has not been empirically validated. Every claim in this paper is architectural design, not experimental result. The benchmark results that would convert design into evidence are the immediate next step.

The authority router does not yet exist as code. The trace library does not yet contain validated traces. The generation pipeline has not been run. No coordinator has reviewed any trace for clinical accuracy. These are not minor qualifications—they are the gap between architecture and artifact.

The 70/25/5 volume split across tiers is estimated, not measured. The actual distribution depends on the coordinator population's query patterns, which can only be determined through deployment. If the proportion of edge and full-mesh queries is higher than estimated, the compiled tier handles less than projected.

The multi-agent debate convergence criteria are underspecified. The multi-turn token budget across case lifecycles needs investigation. Attention dynamics in large context windows may affect primitive and trace uptake. The AOPO custodian proposal for community governance is aspirational. These are the research agenda for the next phase.

The self-obsoleting property assumes that the regulatory domain is finite and that the trace library can asymptotically cover it. If regulatory complexity grows faster than traces can be generated and validated, the system may not converge. In practice, the bounded nature of the OPO regulatory environment makes convergence likely, but this remains an empirical question.

14. Conclusion

The organ procurement community needs AI systems that reason correctly about regulatory compliance under the most demanding conditions: complex multi-authority intersections, time-critical decisions, consequences measured in human lives. This paper identifies the deepest architectural insight of the AORTA research program: for the vast majority of these decisions, the reasoning is already fully specified by the regulatory structure itself. The language model is an expensive interpreter simulating a computation that can be compiled and executed directly.

The compiled core transforms the language model from a runtime into a factory. The factory manufactures two artifacts—a compositional grammar and a reasoning trace dataset—that together provide the complete reasoning methodology for organ procurement regulatory compliance. A three-tier runtime dispatches queries to the minimum computational tier required: deterministic lookup for routine queries, grammar-guided generation for intersections, full compositional reasoning for genuinely novel scenarios. The compiled tier requires no AI, no API, no GPU, and no internet. It runs on commodity hardware in microseconds with zero hallucination risk.

The system gets safer over time because it gets more deterministic. Each validated trace adds to the compiled library. Each promoted query reduces the LLM's runtime role. The architecture is explicitly designed to eliminate its own need to use AI—and that elimination is what makes it trustworthy in healthcare.

The framework is portable across models, organizations, and regulatory updates. It is released under open license. Any OPO can adopt the compiled tier immediately, regardless of technical sophistication, and expand to higher tiers as infrastructure permits.

We invite the OPO community to participate in building the shared trace library, validating the authority router, and running the empirical experiments that will convert this architecture into evidence. The design is complete. The implementation begins now.

The compiled core architecture is not dependent on any particular language model or generation pipeline. Any sufficiently capable system can manufacture the artifacts. As language models improve, the quality of generated traces improves, and the generation pipeline becomes more efficient. But the compiled artifacts themselves are model-agnostic. An OPO that builds its Tier 1 appliance using traces generated by today's models can continue using that appliance indefinitely, even if tomorrow's models are radically different. The architecture decouples the rate of AI progress from the rate of deployment. The compiled core is a frozen asset in a flowing stream.

Every primitive composed is an intersection mastered. Every trace compiled is a hallucination eliminated. Every query that runs in Tier 1 is a coordinator freed from waiting for a language model to simulate a computation that was already solved.

Every organ saved is a life continued.

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